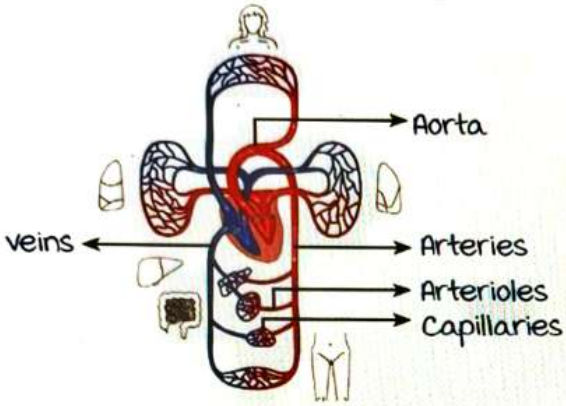
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CHARACTERISTICS OF BLOOD VESSELS

----- Active space ---

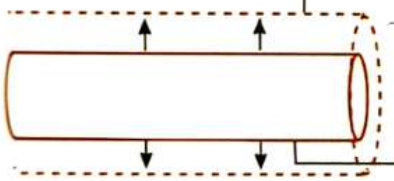


Aorta

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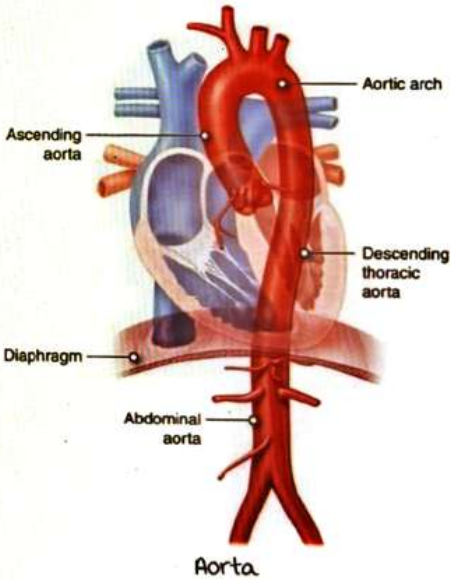
ELASTICITY/WINDKESSEL EFFECT :

- Maintains diastolic blood pressure.
- D/t : Elastin & fibrillin (Supporting protein).

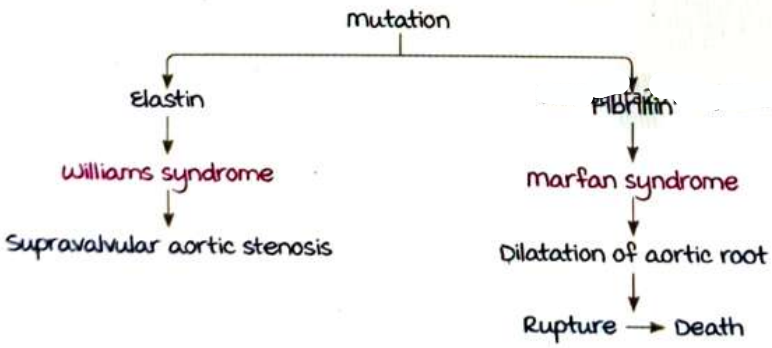


During systole : Stretches → SBP = 120 mmHg

During diastole : Undergoes elastic recoil → DBP : 80 mmHg



Applied aspect :



VELOCITY & CROSS SECTIONAL AREA :

$$\text{velocity of blood flow} \propto \frac{1}{\text{cross-sectional area of blood vessel}}$$

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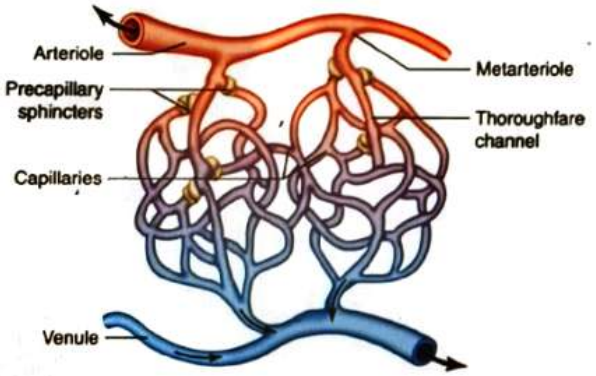
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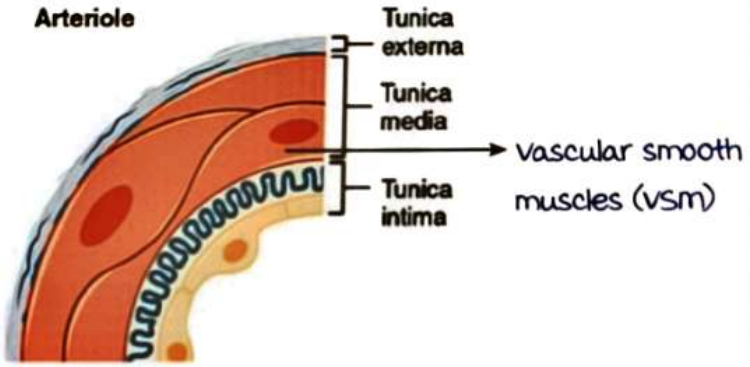
- Aorta :
  - Max. velocity → Thickness & blood pressure.
  - Lowest cross-sectional area.

Arterioles

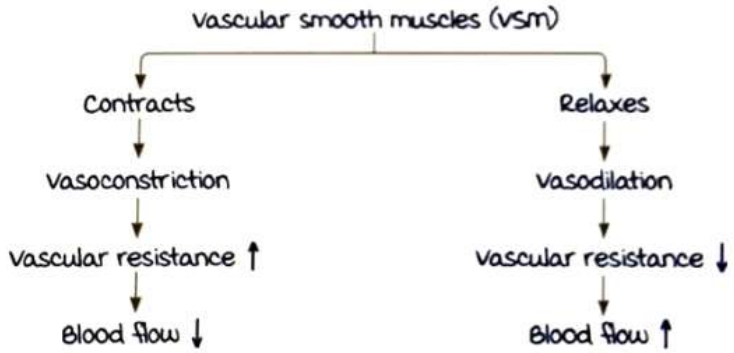
00:08:58

- AKA resistance vessels.





- Function : Regulates blood flow.

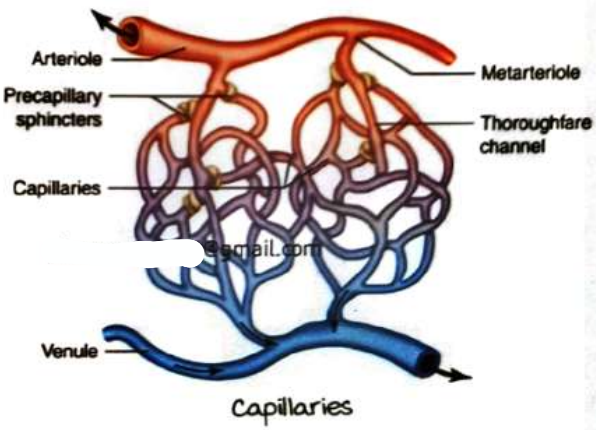


Capillaries & lymphatic system

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CAPILLARIES :

- AKA exchange vessels.
- Function : Exchange of nutrients & gases.

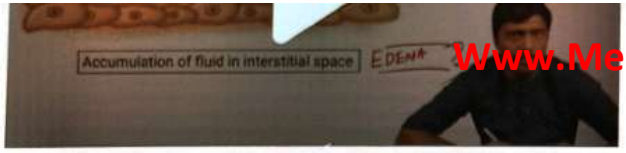


Capillaries

Characteristics :

1. Volume of blood in capillaries: 5% of total blood.
2. Flow of blood: Intermittent.
   - D/t intermittent contraction & relaxation of precapillary sphincters (Vasomotion).
3. Cross-sectional area : Highest.
4. Blood flow velocity : Lowest (Sluggish).

Feedback



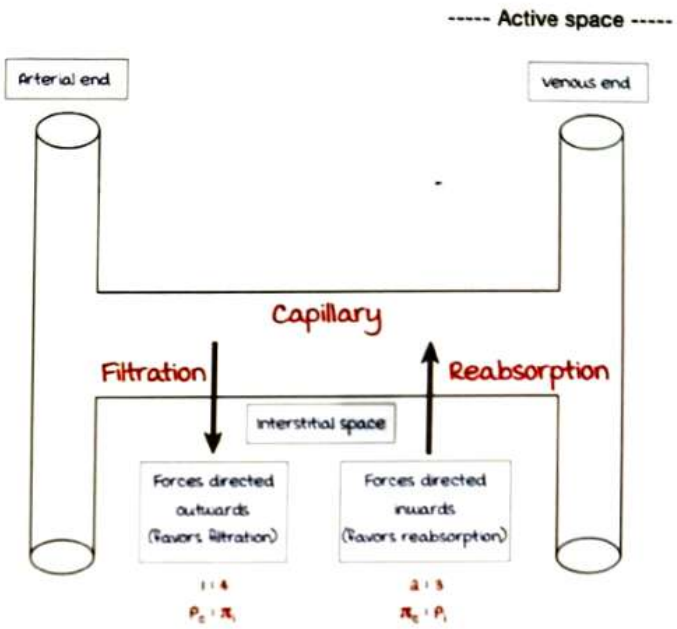
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Starling forces :

1. Capillary hydrostatic pressure (P_c) = 25 mmHg.
2. Interstitial oncotic pressure (π_i) = 3 mmHg.
3. Capillary oncotic pressure (π_c) = 25 mmHg.
4. Interstitial hydrostatic pressure (P_i) = −2 mmHg.

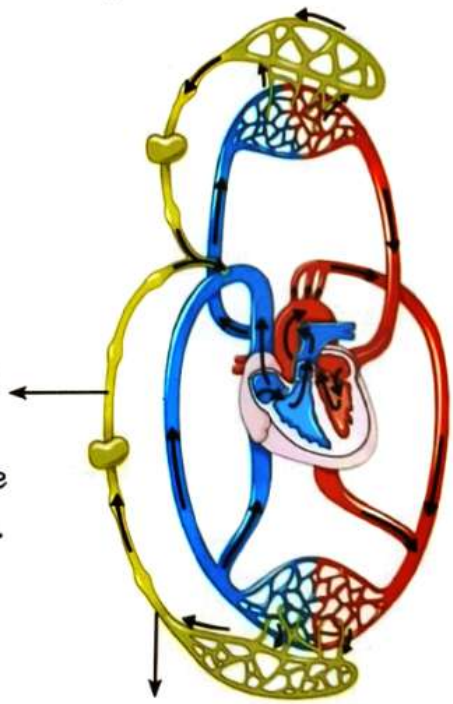
Net filtration pressure
$= (P_c + \pi_i) - (\pi_c + P_i)$

Note: P_i = Positive in : Skeletal muscle, liver & brain.



Lymphatic circulation :

- Filtration >> reabsorption.



Carries excess fluid from interstitial space to blood vessels.

lymphatic vessels

Factors ↑ lymph flow :

1. Lymphatic pump.
2. Muscle contraction.
3. Blood vessel pulsations.
4. Massaging.
5. ↑ Pc (into physiological limits).

Applied aspect:

Edema :

- Accumulation of fluid in interstitial space.
- Causes :

1. ↓ π_c : ↓ plasma proteins
   - ↓ production
     - Liver disease (Eg : Cirrhosis)
   - ↑ excretion
     - Nephrotic syndrome
2. ↑ Pc (Beyond physiological limit) :
   - Chronic hypertension → Lymphatic regulation fails.
3. Lymphatic obstruction
   - Filariasis.

Feedback

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----- Active space -----

Veins

00:34:12

AKA capacitance vessels.

Characteristics :

- Volume of blood in veins = 55% of total (Storage reservoirs).
- High distensibility/compliance :

↑↑ Compliance = ↑↑ ΔV (Change in volume) / ↓↓ ΔP (Change in pressure) } High volume low pressure system

Applied aspect :

Lying down → ↑ Venous return → Stored in Pulmonary veins

↓

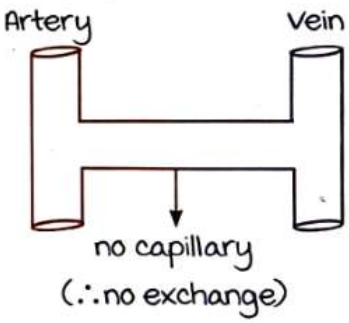
Pulmonary congestion

↓

Breathlessness (Orthopnea)

Note :

- Vena cava :
  - max diameter.
  - min blood pressure.
- Arteriovenous anastomosis/shunt vessel :
  - Sites : Fingertips, ear lobes.
  - ↑ in : Temperature regulation.



Feedback